

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. PRODUCTION MACHINE LEARNING MODEL PIPELINE (train.py):

The predictive algorithm training loop structure uses Scikit-learn's Random Forest architecture to parse non-linear environmental relationships:

```
```python
import pandas as pd
import pickle

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier

Load Matrix
data = pd.read_csv('data/crop_data.csv')
X = data[['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']]
y = data['label']

Split Data Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

Fit Estimators
model = RandomForestClassifier()
model.fit(X_train, y_train)

Export Serialized Engine Binary
with open('models/crop_model.pkl', 'wb') as f:
 pickle.dump(model, f)
```